

Customer Participation in Lead Pipe Replacement: A New Jersey Case Study

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Key Takeaways

- Water systems require customer participation to meet federal requirements for inventorying and replacing drinking water pipes if they contain lead
- Renter-occupied and lower-value properties were less likely to participate in a voluntary lead pipe replacement program in New Jersey
- An assessment of two grant programs subsidizing low-income homeowners' replacement costs found that only the well-publicized and easy-to-access program boosted participation
- Results showed that, even with subsidies, it was difficult to get participation in the voluntary program among owners of properties with lead pipes

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Introduction

Exposure to lead is a critical public health issue in the United States that is linked to children's learning and behavior problems, among other health disorders (EPA 2024). No safe level of lead exposure has been identified (CDC 2024). Drinking water remains a primary pathway for lead exposure, particularly for infants (Zartarian et al. 2017).

Lead service lines (LSLs), the pipes that supply drinking water from water mains to homes, are the largest contributor to water lead levels nationally (EPA 2019). Recent federal regulations require nearly all water systems to replace LSLs under their control by 2037 (EPA 2024). While federal law banned the installation of new LSLs in 1986, EPA's recent inventory of lead service lines, nationally, dropped the number of such lines from its prior estimate of 9 million to 4 million including "unknowns" (EPA 2025). These pipes pose an increased risk of lead exposure to consumers (Brown et al. 2011; EPA 2024; Triantafyllidou and Edwards 2012).

Customer involvement is vital to the success of LSL replacement programs because inspection often requires access to the inside of buildings. Unlike many other types of publicly managed infrastructure, service line ownership in many U.S. communities is split between the water system and the property owner, making property owner participation necessary for the water system to replace the privately owned portion of the water line.

Water systems rarely have the authority to conduct a LSL replacement without a signed consent form or right of entry. Replacement is also costly, running several thousand dollars per property. Even when full subsidies are offered, non-financial barriers can dampen customer participation in both inspection and replacement.

To provide insights into effective program design, Klemick et al. (2024) conducted an analysis of the characteristics of customers who participated in a voluntary service line inspection and replacement program in a large water system located in Trenton, New Jersey. They also evaluated the program features that yielded the highest participation rates. The authors found that LSLs were more prevalent in the urban municipality of the water system, which has a higher concentration of older housing stock, Black and Hispanic residents, renters, and poverty than suburban municipalities in the service area. However, the study also found that renter-occupied and lower-valued properties were less likely to participate in LSL program registration, inspection, and replacement, indicating potential barriers to participation. The analysis showed that grant programs available to subsidize LSL replacement costs for certain properties

did not increase LSL replacement rates unless they were both easy to access and accompanied by substantial outreach.

Barriers to Customer Participation

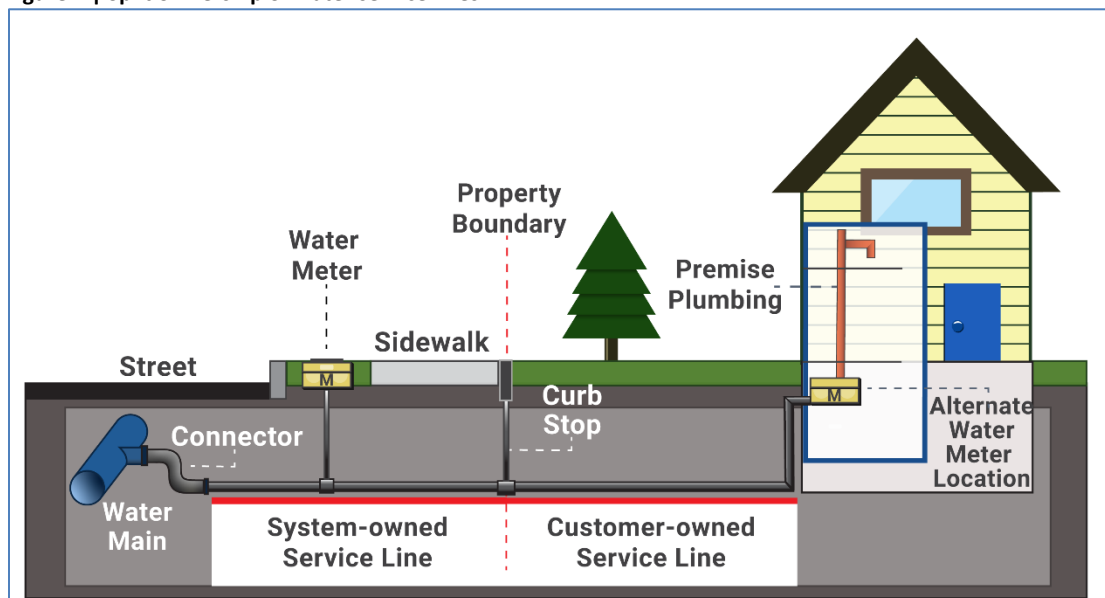
Inspection and replacement of customer-side LSLs is challenging due to numerous barriers. Financial barriers are relevant in systems that do not fully cover the costs of replacing the homeowner's side of the line. Some utilities in the United States may face state or local laws requiring property owners to cover at least some portion of replacement costs (EPA 2024). EPA (2024) estimated customer-side LSL replacement construction costs to range from around \$1,900 to \$5,400, with other sources citing costs up to \$10,000 (American Water Works Association 2022).

Non-financial barriers include:

- lack of awareness about the risks of lead exposure
- lack of trust in water systems
- language barriers
- the time and inconvenience of scheduling an inspection or replacement
- disruption to one's home or yard

Furthermore, ownership of the service line is typically split, with the utility owning the portion from the water main to the property line, and the property owner owning the remainder (Figure 1). Consequently, utilities may require property owner permission for inspection and replacement. Rented properties can present a barrier by requiring additional layers of communication, coordination, and outreach between the water system, property owner, and renter. These barriers are often more pronounced for low-income residents, and some were exacerbated during the COVID-19 pandemic, further decreasing households' willingness to allow in-person inspections and construction.

Figure 1 | Split ownership of water service lines



This illustrative service line configuration shows the typical split ownership of the service line between homeowner and utility. Either or both sides can contain lead. Public water systems may have records of where lead-containing utility-owned service lines are located. Inventory on the customer-side of the service line is typically less well documented and is a necessary step in replacing water service lines. Source: EPA 2022.

Utilities encouraging service line inspection and LSL replacement often focus on improving outreach and reducing costs (American Water Works Association 2005; EPA 2019). This can include providing discounted upfront pricing, low or no interest financing, and translating communication materials into languages other than English. Systems in some cities, such as Washington, D.C., and Providence, Rhode Island, have targeted free or subsidized replacements to low-income homeowners or properties in disadvantaged neighborhoods (EPA 2019; Kuffner 2022). The state of Michigan requires most water systems to finance the full cost of LSL replacement for all properties, including the customer side (EPA 2019); costs are recouped through a combination of grants, bond financing, and rate increases spread over all utility customers. While voluntary participation is typical, some cities require replacements for confirmed LSLs and fully cover the costs. For example, Newark, New Jersey, adopted a free and mandatory approach including fines for non-compliance (City of Newark 2019).

Lead Service Line Replacement in Trenton, New Jersey

Trenton Water Works serves more than 60,000 properties in Trenton and four neighboring municipalities. It launched an LSL replacement program in 2019 in response to exceedances of EPA's action level for drinking water lead concentrations. Trenton Water Works offered replacement of customer-side LSLs

(including lead-lined galvanized steel) for a subsidized cost of \$1,000 payable over five years without interest as a special tax assessment on the property, with utility-side lines replaced at the same time at no extra charge when they contained lead.

The utility used various outreach approaches to encourage program registration, inspection, and replacement, including community meetings, door-to-door visits, and mailings. Registration for the program did not entail a formal commitment by the property owner to pay the \$1,000 cost and proceed with inspection – it was only a signal of interest. However, if the customer side was confirmed to be lead, the property owner had to sign a replacement agreement committing the property to the special tax assessment to have the line replaced. By the end of 2022, Trenton Water Works had conducted more than 17,000 service line inspections and replaced more than 9,000 lines found to contain lead, including 2,500 customer-side lines.

Klemick et al. (2024) examined the characteristics of properties and neighborhoods more likely to have LSLs and to participate in the Trenton Water Works inspection and replacement program during 2019-2022. Customer-level data from the water system was linked to property characteristics from the county's property tax assessment database and neighborhood characteristics from the 2020 U.S. Census to understand how LSL location and participation in the LSL replacement program varied across properties and neighborhoods. Klemick et al. (2024) provides more details about these analyses.

The analysis also evaluated the impact on service line inspection and LSL replacement of two different grant programs introduced in 2021. Both programs offered full-cost subsidies to certain residents in Trenton. The first grant program was run by a community-based organization and targeted residents of a particular low-income neighborhood in Trenton. This grant covered the full \$1,000 LSL replacement cost for owner-occupied homes and \$500 of the cost for renter-occupied homes, with landlords covering the remainder. The grants required minimal paperwork from applicants, and the community organization leveraged its communication channels, conducting extensive outreach through in-person and virtual events, door-to-door and neighborhood canvassing, and social media to notify and educate residents in the target neighborhood about the program.

The second grant program was offered by Trenton's Department of Housing and Economic Development, funded by a U.S. Department of Housing and Urban Development Community Development Block Grant, to subsidize urgent home repairs for low- and moderate-income homeowners. LSL replacement was added as a qualifying home repair in 2021, meaning that the entire \$1,000 cost to customers would be

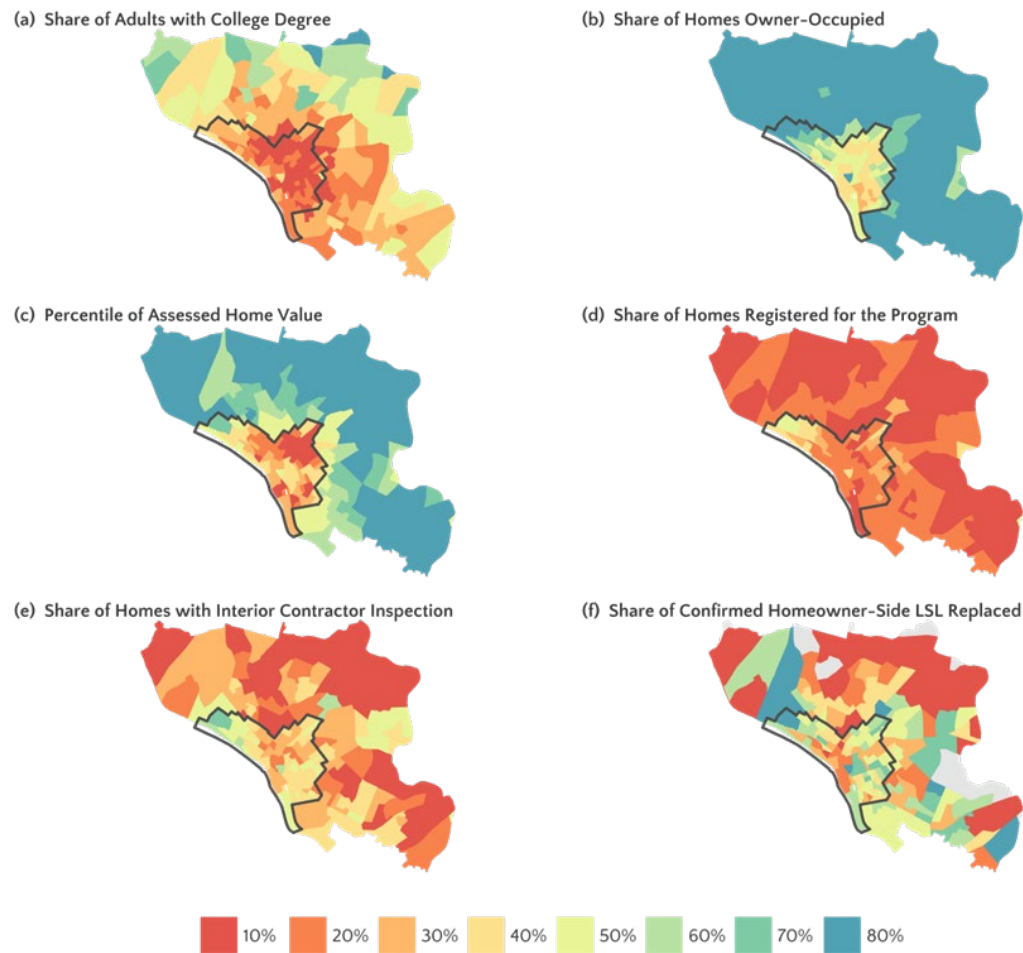
covered for successful applicants. Applicants were required to provide extensive documentation to prove their eligibility, such as wage and withholding statements and tax returns.

Characteristics of LSL Location and Customer Participation

The analysis found that, prior to any replacements, LSLs were more prevalent in the urban municipality than the suburban municipalities. The urban municipality had a higher concentration of older (pre-1951) housing, Black and Hispanic residents, renters, and poverty than the suburban municipalities in the service area. Because of the higher concentration of LSLs, the utility sent more outreach letters to urban properties to encourage LSLR program registration than it did to suburban properties.

Figure 2 shows the spatial distribution of property, neighborhood characteristics, and program participation in the study area. Consistent with the higher concentration of LSLs, urban properties were more likely to register for the program, have an inspection by a contractor, and replace confirmed customer-side LSLs than suburban properties. Urban properties were also more likely to be renter-occupied, have lower assessed values, and be in neighborhoods with lower rates of college graduation.

Figure 2 | Spatial distribution of neighborhood and property characteristics and LSL replacement program participation



Individual LSL replacement program participation and property assessor data on tenancy and home values were aggregated to the Census block group level. Census block groups are included if they fall within the service boundary of the public water system and are part of one of four municipalities where the water system conducted the LSL replacement program. The interior of the black outline denotes Trenton, the urban municipality, and the remaining areas comprise the suburban municipalities in the study area. Gray shading in panel (f) denotes block groups where LSL replacement is not applicable because there were no confirmed customer-side LSLs.

Separate analyses of the urban and suburban municipalities were conducted to better understand program participation trends. The full results in Klemick et al. (2024) show that owner-occupied properties, higher valued properties, and single-family homes were all more likely to participate in service line inspections and replacements of confirmed lead lines in both urban and suburban municipalities. For example, replacement rates among urban properties with confirmed LSLs were 21 percentage points higher in owner-occupied properties than in renter-occupied properties. In suburban municipalities, owner-occupied properties were 7 percentage points more likely to replace their LSLs than renter-occupied properties.

Properties with higher assessed value were significantly more likely to register and to have an inspection in both urban and suburban municipalities, and significantly more likely to replace a confirmed LSL in the urban municipality than properties in these areas with lower assessed values. These results suggest that there are substantial barriers to LSL replacement for rental properties and low-income households. However, Klemick et al. (2024) were unable separately identify the role of financial and non-financial barriers since low-income households and renters disproportionately face both types of barriers.

The analysis found that neighborhood socioeconomic characteristics such as race, ethnicity, and share of college graduates had inconsistent associations with service line inspection and LSL replacement program participation. Registration rates were sometimes higher in areas with a higher share of Black or Hispanic residents, suggesting that non-financial barriers such as lack of information, mistrust, and language barriers were not necessarily higher in these neighborhoods. However, registration rates were also higher in neighborhoods with a higher share of college graduates. Ultimately, participation in replacement of confirmed LSLs was lower in neighborhoods with a higher share of Black residents, which could reflect additional barriers not reflected in rental status and property values.

In addition, properties whose utility side of the service line was lead and those sent more outreach letters by the utility were more likely to participate in service line inspection and LSL replacement. Properties located on blocks with a street construction moratorium were also less likely to participate. These results are consistent with direction provided by the utility to the contractors about where to focus inspection and replacement efforts.

Grants Don't Always Boost Customer Participation

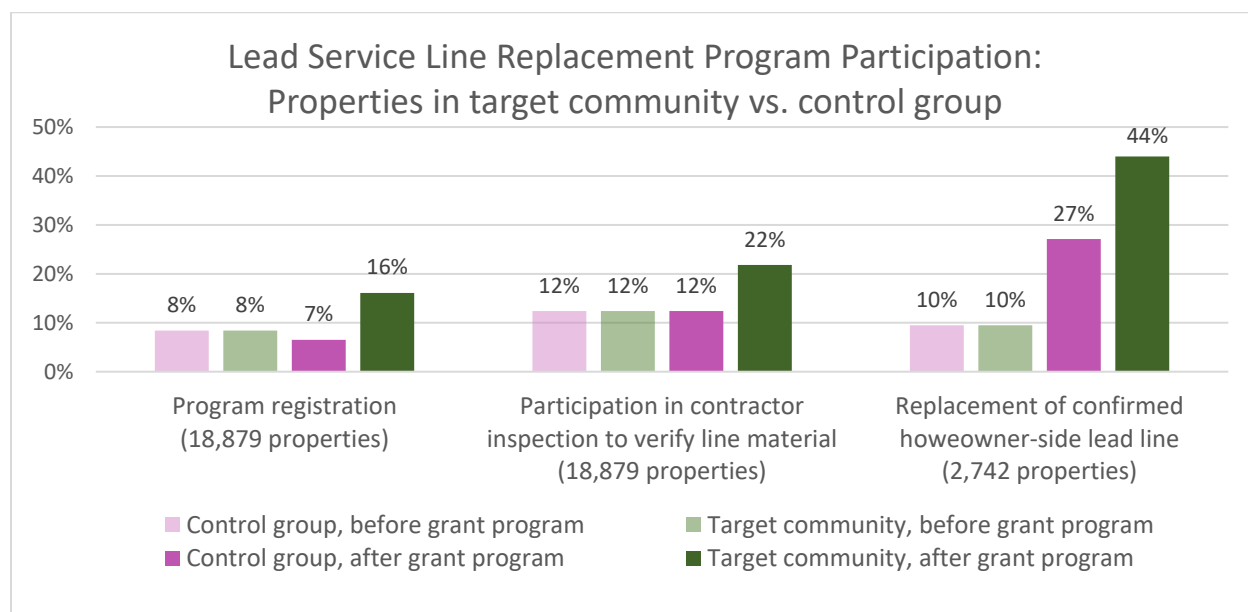
Klemick et al. (2024) estimated the effects of the community-based grant program using statistical techniques to identify a control group of properties that had similar characteristics as those in the target community but were located in other neighborhoods. The authors used regression analysis to examine program participation before and after the grant program was launched in the target community compared to the control group. Figure 3 presents the results of this analysis.

Before the grant was introduced in 2021, rates of program registration, inspection, and replacement of confirmed lead lines were virtually identical in the target community and the control group (Figure 3, green vs. purple light-shaded bars). After the grant program was introduced, LSL program participation in the target community was substantially higher than in the control group (green vs. purple darker-shaded

bars). Registration rates were twice as high, and inspection and replacement rates were more than 50% higher in the target community than in the control group, differences that are statistically significant.

LSL replacements also increased significantly in the control group after the grant program was introduced, even though they did not rise as much as they did in the target community. An increase in the rate of LSL replacements occurred over time throughout the study area as contractors gained experience and increased their capacity to implement replacements, and as residents' awareness of the program grew. The rate of replacements was also higher in the later period because restrictions related to the COVID-19 pandemic had been lifted.

Figure 3 | The effect of community-based grant program on LSLR registrations, inspections, and replacements



To evaluate the housing department grant program available to low-income homeowners in Trenton for urgent home repairs, the water utility worked with the research team to send postcards to a randomly selected group of potentially eligible customers informing them about the program (Figure 4). These properties were then compared to a control group of customers located in the same neighborhoods who were also potentially eligible for the grants but were not sent postcards.

Figure 4 | Postcard sent to randomly selected, potentially eligible properties



Take advantage of this special opportunity and **REPLACE YOUR LEAD PIPES!**

Why should I replace my lead service line?

Exposure to lead in drinking water may cause serious health problems including damage to the kidneys and brain. **Infants, children, and those who are pregnant are especially at risk.**

CITY OF TRENTON INCOME ELIGIBLE HOMEOWNERS MAY QUALIFY FOR GRANT ASSISTANCE

Qualifying homeowners can have their entire lead service line (including the portion they own) replaced at **NO COST** through the Trenton Urgent Rehabilitation Program (TURP)!

HOW TO APPLY:

STEP 1

Check if your home has lead pipes by visiting Trenton Water Works (TWW) at **www.twwleadprogram.com** and clicking "check your line," or by contacting TWW's Lead Service Line Replacement Program: **(609) 989-3600** or **twwleadprogram@trentonnj.org**

STEP 2

Submit an application to TURP for financial assistance. Applications and eligibility criteria are available at **www.trentonnj.org/turp** or at City Hall, 319 East State Street, Trenton NJ. Select "Lead Service Line Replacement" on the application. The City will provide an award letter if you are eligible for the grant.

STEP 3

A contractor will then contact you to schedule an appointment for free lead pipe replacement. JAS Group is the contractor hired by TWW to conduct replacements in this special program.

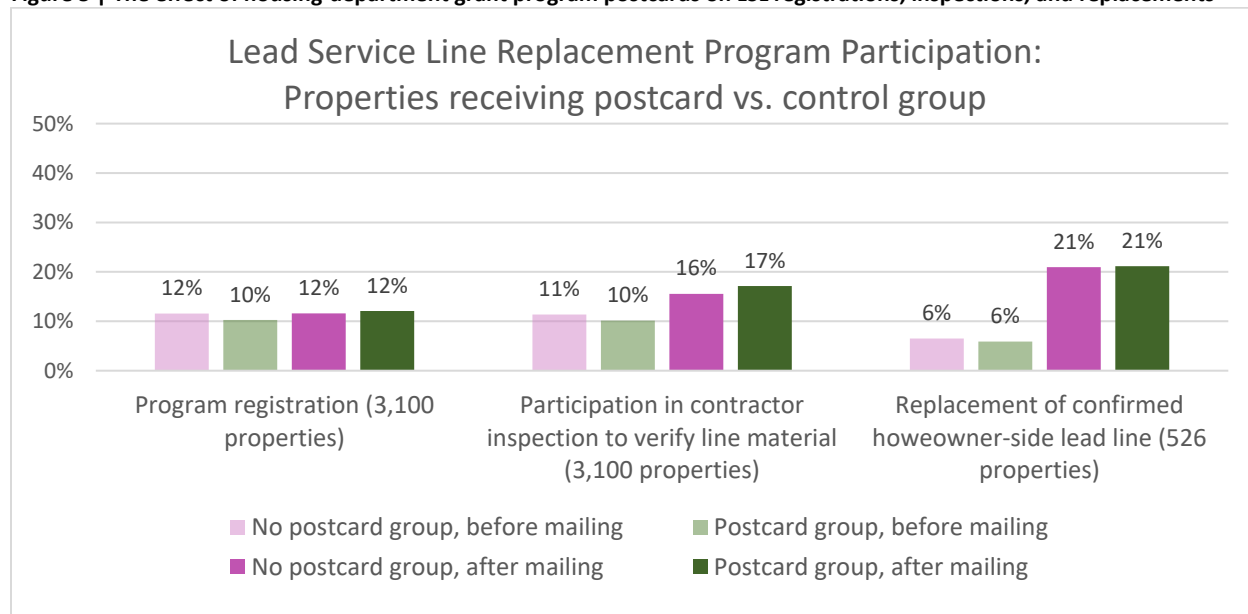
Property owners are responsible for maintenance of their water service line from the curb to inside their house. Trenton Water Works is responsible for the water service from the main in the street to the curb. Both sides can contain lead.

Figure 5 shows the results of the postcard experiment promoting the housing department grant program. The analysis found that program registration, inspection, and replacement of confirmed LSLs were similar across the postcard and no-postcard groups before the intervention (green vs. purple light-shaded bars). After the postcards were sent, there was an increase in all three program participation measures in both the postcard and no-postcard groups, consistent with the increase in LSL replacement program activity during the study period throughout the service area (green vs purple darker-shaded bars). There were slightly larger increases in registration and inspections in the postcard group than the no-postcard group, and no differences in the replacement rates of confirmed LSLs across the two groups. None of the differences across the two groups were statistically significant, indicating no meaningful difference in participation. The increase in replacement rates among properties with confirmed customer-side LSLs was nearly identical across the two groups. Furthermore, housing department staff confirmed that they did not receive any grant applications for LSL replacement after the postcards were sent. These results suggest that the postcard about the grant opportunity was ineffective in boosting LSL replacement program participation.

Because the postcards only included English-language information, they could have been less effective in areas with a higher concentration of non-English speakers. However, an additional analysis revealed that

the effect of receiving a postcard did not vary based on the share of Hispanic residents in the neighborhood.

Figure 5 | The effect of housing-department grant program postcards on LSL registrations, inspections, and replacements



These results yield insights about what approaches motivate participation among customers that might face a combination of barriers. The results from the postcard experiment show that providing information about subsidies to cover LSL replacement costs was not sufficient to attract customer participation. However, the different results in response to the community-based program suggest that subsidies were more effective when combined with outreach and trusted community messengers to address non-financial barriers to participation.

The two programs differed substantially in their modes of outreach and application processes, which are factors relevant to non-financial barriers. The community-based grant program was accompanied by extensive outreach and education about LSL replacement, while the experiment promoting the housing department grant program involved a single postcard, and the program was not otherwise well publicized. In addition, the housing department grant program required an application including formal documentation of income to prove eligibility, while the community program had no income requirements. The different results between the two programs suggest that grant programs that address financial barriers alone are not sufficient to boost participation.

Furthermore, participation in the neighborhood targeted by the community grant program, while much higher than in the control group, topped out at 54% of properties with confirmed LSLs during the study period even when the financial barrier to participation was removed for homeowners and substantially reduced for landlords. A representative from the community organization confirmed that the program could have funded more LSL replacements had they received more applications, indicating that the shortfall was due to lack of customer demand rather than insufficient funding. Considering that only 41% of residential properties in the target neighborhood were owner occupied and that only these properties were eligible for full \$1,000 subsidies, it is possible that the grant program could have had a larger effect if rental properties had also qualified for full rather than partial LSL replacement subsidies.

Conclusions

The results from Klemick et al. (2024) confirm that water systems designing programs to identify and replace LSLs need to address a range of financial and non-financial barriers to achieve the goal of replacing all LSLs. The findings suggest that simple mailers providing information to households on how they can apply for assistance are not an effective way to boost participation. They also suggest that addressing financial barriers alone—even when those barriers are sizable—is not enough to motivate participation among low-income customers. Rather, the findings suggest that programs that are free, well publicized, and easy to access are better at encouraging customers to voluntarily participate in programs to identify and replace their lead service lines.

Another conclusion from Klemick et al. (2024) is that even free programs with well-designed outreach face substantial challenges to achieve full participation when replacement is voluntary. In the face of these types of challenges, the City of Trenton and other municipalities in the service area made LSL replacement both free and mandatory in 2024 and rescinded the special property tax assessments levied to finance LSL replacements (City of Trenton 2024a, 2024b). More case studies chronicling water system experiences can help identify other best practices for lead service line identification and replacement, even if there is no “one size fits all” solution for communities confronting large numbers of legacy lead pipes.

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Conflict of interest statement

The authors declare they have no financial competing interests. Non-financial competing interests are declared in the authors' cover page.

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